When Averaged Field Regularity Fails to Rank Few-Step Generative Paths

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Abstract

Choosing a probability path for few-step generative sampling is often guided by averaged regularity of the velocity field: a path with smaller averaged squared Jacobian norm is expected to discretize better at a fixed number of function evaluations (NFE). We test this assumption where every quantity is exact: commuting Gaussian probability flows, whose numerical endpoint laws and Gaussian 2-Wasserstein (W_2) errors are closed-form. The averaged-regularity ordering of linear versus variance-preserving paths fails to determine the fixed-NFE W_2 ordering: fourteen ranking inversions reproduce across Euler, Heun, and RK4, survive an 80-digit precision audit, and recur in a pre-registered non-centered family. An exact modal decomposition explains the mismatch through solver-stage sampling and signed one-step defects transported to the endpoint, and an explicit scalar construction proves the non-implication for fixed-grid Euler. This is a controlled limitation study: it does not refute regularity-guided schedule design, and no learned model is evaluated.

1 Introduction

Few-step sampling of continuous-time generative models commits to a probability path, a solver, and a small NFE budget before any error is observed. A natural criterion, made explicit by Lipschitz-guided schedule design [Chen et al., 2025], ranks candidate paths by the averaged squared Jacobian (Lipschitz) norm of the velocity field. Averaged regularity controls worst-case error *bounds* [Hairer et al., 1993], but bounding an error is weaker than determining which of two paths errs less at an equal small budget. We isolate that question where the marginal flow is affine, the fixed-step endpoint law is exactly computable, and the error is the closed-form Gaussian W_2 [Gelbrich, 1990]: no sampling, training, or estimation noise intervenes, and every reported digit is auditable.

Contributions.

- A controlled exact-W_2 benchmark on commuting Gaussian flows in which the averaged-regularity ordering fails to determine the fixed-NFE W_2 ordering in 14 of 36 pre-specified comparisons, verified against an 80-digit reference and reproduced (11 of 18 comparisons) in a pre-registered non-centered family frozen before execution.
- An exact eigenmode analysis showing the endpoint factor error is a transported sum of signed, solver-stage-dependent local defects—information an unsigned time average of Jacobian norms cannot retain.
- An explicit smooth scalar construction proving that, for every $L > 0$ and grid size N, strictly larger averaged squared Jacobian is compatible with zero fixed-grid Euler endpoint error.

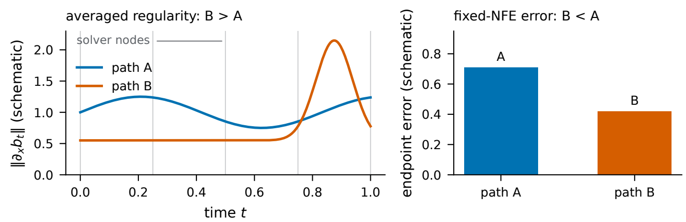

Figure 1: **Conceptual schematic (not data).** Path B has the larger time-averaged Jacobian magnitude, but its irregularity concentrates between the nodes a fixed-NFE solver samples; the endpoint-error ordering can oppose the averaged-regularity ordering.

Solver dependence of good schedules is not itself novel [Sabour et al., 2024, Karras et al., 2022]; the contribution is the exact, controlled non-implication and its modal explanation, which bound what averaged regularity can certify about fixed-budget *rankings* in the tested systems.

2 Gaussian probability flows and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent and let a scalar schedule $(\alpha(t), \sigma(t))$, with $\alpha(0) = \sigma(1) = 1$ and $\alpha(1) = \sigma(0) = 0$, define the interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023]. The marginal law is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q(t) = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$, and the marginal velocity field is affine,

$$b(t, x) = A(t)\, x + c(t), \qquad A(t) = C_t\, Q(t)^{-1}, \qquad c(t) = \dot m_t - A(t)\, m_t, \tag{1}$$

with $C_t = \dot\alpha\alpha\, \Sigma_0 + \dot\sigma\sigma\, \Sigma_1$; centered with $\Sigma_0 = I$ this is $A(t) = \frac{1}{2}Q'Q^{-1}$, $c \equiv 0$. We compare the *linear* path $(1 - t,\, t)$ and the *variance-preserving* (VP) path $(\cos\frac{\pi t}{2},\, \sin\frac{\pi t}{2})$ [Song et al., 2021]. Following Chen et al. [2025], the averaged-regularity baseline is $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2\, dt$ (exact spectral norms of a smooth closed-form integrand, fixed trapezoidal grid); by its own definition $\mathcal{R}$ ignores $c(t)$. We measure error with Gaussian W_2 because it is the exact, closed-form, distribution-level metric used by this schedule-design literature.

A fixed-step solver applied to (1) induces an affine endpoint map, recovered exactly from $d+1$ probe trajectories; pushing the source moments through it gives the exact endpoint error [Gelbrich, 1990]. With $\Sigma_0 = I$ and $\Sigma_1 = U\mathrm{diag}(\lambda_i)U^\top$ all matrices commute: each mode obeys $x_i' = a_i(t)x_i + c_i(t)$ with $a_i = q_i'/(2q_i)$, $q_i = \alpha^2 + \sigma^2\lambda_i$, and centered, $W_2^2 = \sum_i(|r_i| - \sqrt{\lambda_i})^2$ with r_i the product of the numerical one-step factors of mode i.

3 Why the ranking need not agree: transported signed defects

The exact one-step transition of mode i over $[t_n, t_{n+1}]$ is $e_{i,n} = \sqrt{q_i(t_{n+1})/q_i(t_n)}$, while a Runge–Kutta step multiplies by a factor $r_{i,n}$ built from stage samples of a_i (one for Euler, two for Heun, four for RK4). For smooth a, the leading local defect (exact minus method) is $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-\frac{1}{12}a'' + \frac{1}{6}a^3)h^3$ for Heun—different derivatives, signs, and stage locations. The endpoint factor error obeys the exact telescoping identity

$$\prod_{n<S} r_{i,n} - \prod_{n<S} e_{i,n} = \sum_{n<S} (r_{i,n} - e_{i,n}) \Big(\prod_{j<n} r_{i,j}\Big) \Big(\prod_{j>n} e_{i,j}\Big), \tag{2}$$

with no smallness or positivity assumption. This identifies what the scalar $\mathcal{R}$ discards: which times the stages sample, the derivatives of A, the *signs* of local defects, their cancellation, and their transport to the endpoint. An unsigned average of $\|A\|^2$ retains none of these, so it need not determine the fixed-NFE ordering.

4 Controlled benchmark and results

Design. Source $N(0, I)$; anisotropic (condition number 4) and seeded low-rank ($FF^\top + 0.05I$, rank 2) targets; $d \in \{2, 8\}$; linear and VP paths; Euler, Heun, RK4; NFE $\in \{8, 16, 32\}$ with equal-NFE

Table 1: Strongest inversion: low-rank Gaussian, $d = 8$, Euler, NFE 8. (`phase4_gaussian_reproduction_2026-07-24-v1:results`)

path	$\mathcal{R}[b]$ (lower = preferred)	Gaussian W_2
linear	**2.9476523251**	0.8108540111
variance-preserving	4.7295206355	**0.4564779075**

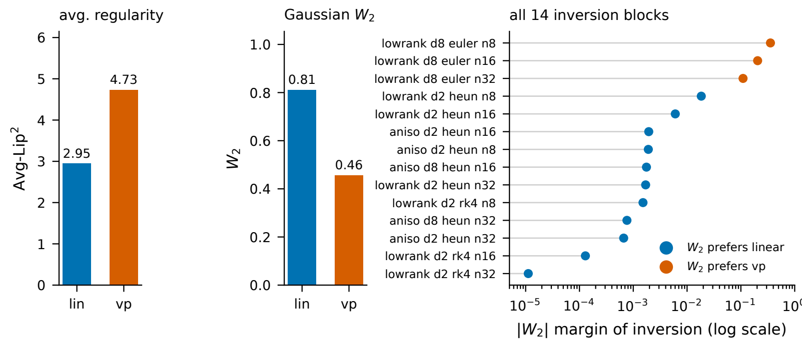

Figure 2: **Left:** the strongest inversion in both quantities. **Right:** W_2 margins of all fourteen inversion blocks (log axis; compresses, never exaggerates); color marks the path exact error prefers, always opposite to the baseline. (`phase4_gaussian_reproduction_2026-07-24-v1:results`)

accounting (steps = NFE/stages): 72 float64 CPU configurations in 36 two-path blocks. A *ranking inversion* is a block where the $\mathcal{R}$ and W_2 orderings strictly disagree. Grid, definitions, and tolerances were fixed before execution; post-hoc diagnostics are labeled as such.

Fourteen audited inversions. All 72 configurations reproduce a registered earlier run within 10^{-12}, and 14 of 36 blocks are inversions (Figure 2; `phase4_gaussian_reproduction_2026-07-24-v1:results`); the strongest, low-rank at $d = 8$ with Euler at NFE 8 (Table 1), has the baseline prefer linear while its W_2 is 78% larger (margin 0.3543761036). All 72 float64 W_2 values agree with an 80-digit `mpmath` reference to at most 9.71×10^{-10}; the smallest inversion margin, 1.1189×10^{-5}, exceeds this 11,500-fold (`phase4_precision_2026-07-24-v1:table`).

Solver–path interaction. In the low-rank grid, VP is preferred under Euler and linear under Heun and RK4, at both dimensions and every primary budget (Figure 3); the pattern persists at NFE 64 and 128 and under $\pm 10\%$ target perturbations (`phase4_robustness_2026-07-24-v1:table`). Solver-dependent schedule preference is established [Sabour et al., 2024]; the point is that a solver-independent scalar cannot represent it. A solver-specific leading proxy from Section 3 agrees with 29 of 36 observed preferences versus 22 of 36 for the baseline, but this is post-hoc and in-sample and carries no predictive claim (`phase4_diagnostics_2026-07-24-v1:table`).

Pre-registered external validation. Before execution we froze one non-centered family: source $N(\mu_0, I)$, $\mu_{0,i} = 0.75(-1)^i$; target $N(\mu_1, D)$, $\mu_{1,i} = 1 + 0.25i$, D geometric with condition number 6—so $c(t) \neq 0$ in (1)—with the same paths, solvers, and budgets. Eleven of eighteen blocks are inversions ($\mathcal{R}$ prefers VP, margin 0.7513; exact W_2 prefers linear), all passing the 80-digit audit (max float64 deviation 2.07×10^{-11}; smallest inverted margin 1.42×10^{-6}) (`workshop_external_validation_2026-07-24-v1:results`). This supports the finding only for the tested systems. No mixture target supports any conclusion here; mixture evidence failed estimator calibration and was excluded.

5 A minimal non-implication for fixed-grid Euler

The implication also fails logically, in one smooth scalar dimension.

Proposition 1. *Fix $L > 0$ and an integer $N \geq 1$; let $h = 1/N$ and let forward Euler use the left-endpoint grid $t_n = nh$ on $[0, 1]$. Set $\epsilon_N = N\{e^{L/N} - 1\} - L > 0$ and define $f_k(t, x) = a_k(t)x$*

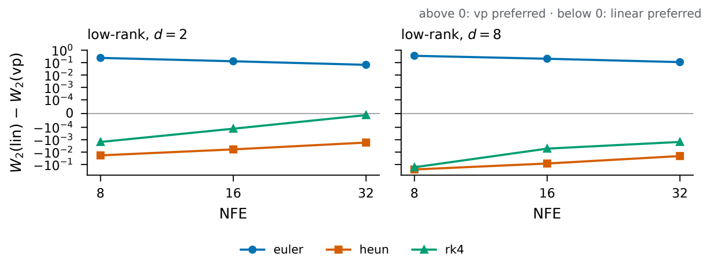

Figure 3: Signed low-rank margins $W_2(\mathrm{lin}) - W_2(\mathrm{vp})$ across solvers and NFE (symmetric log axis): the preferred path flips with the solver at equal NFE. (`phase4_gaussian_reproduction_2026-07-24-v1:results`)

with $a_0(t) = L$ and $a_{1,N}(t) = L + \epsilon_N \cos(2\pi N t)$. For $x(0) = 1$: (i) both exact endpoints equal e^L; (ii) $\int_0^1 a_{1,N}^2 = L^2 + \epsilon_N^2/2 > \int_0^1 a_0^2$; (iii) Euler is exact *for $a_{1,N}$; (iv) Euler has strictly positive endpoint error for a_0. Hence larger averaged squared Jacobian does not imply larger fixed-grid Euler endpoint error.*

Proof. (i) $\int_0^1 \cos(2\pi N t)\, dt = 0$ for integer N, so both coefficients integrate to L and $x(1) = e^L$. (ii) By orthogonality $\int_0^1 a_{1,N}^2 = L^2 + \epsilon_N^2/2$, and $\epsilon_N > 0$ since $e^z > 1 + z$ for $z = L/N > 0$. (iii) At every node $\cos(2\pi N t_n) = 1$, so each Euler factor is $1 + h(L + \epsilon_N) = e^{L/N}$ and the product is e^L. (iv) $(1 + L/N)^N < e^L$ since $\log(1 + z) < z$ for $z > 0$. □

The quantifiers matter: $a_{1,N}$ is chosen after the grid, its amplitude vanishes as $N \to \infty$, and shifted grids or multi-stage solvers sample different phases. This grid-aliasing construction (which passed an independent adversarial review) shows non-implication for fixed-grid left-endpoint Euler only; it is neither the mechanism behind every Gaussian inversion nor a new numerical-analysis phenomenon.

6 Related work

Flow matching and stochastic interpolants define the Gaussian path family and its marginal fields [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023]; Lipschitz-guided schedule design proposes the criterion we test [Chen et al., 2025]. Fast diffusion solvers exploit exactly integrated linear parts and solver-specific expansions [Lu et al., 2022a,b, Zhang and Chen, 2023], EDM separates schedule from discretization [Karras et al., 2022], and Align Your Steps optimizes schedules per solver, model, and dataset [Sabour et al., 2024]: schedule quality is known to be solver-dependent. Runge–Kutta and backward error analysis supply the local-error machinery [Butcher, 1963, Hairer et al., 1993, 2006]; Gaussian W_2 is due to Gelbrich [1990]. None states the exact commuting inversion benchmark or the grid-aware non-implication; search absence does not establish novelty.

7 Limitations and conclusion

Our systems are exact, commuting, affine Gaussian flows with two paths, three solvers, dimensions 2 and 8, and small budgets, plus one pre-registered non-centered family. No learned neural field is evaluated; no image, video, or world-model improvement is demonstrated or implied; the proxy is post-hoc and in-sample; the proposition is grid-aware. Exponential or exact-linear integrators would solve these affine fields exactly, so the result concerns generic fixed-stage Runge–Kutta discretization, not best-possible solvers; and we rank two fixed paths at the identity time parameterization, so reparameterization-based schedule optimization via the transfer formula of Chen et al. [2025] is a complementary use we do not address. Averaged regularity may remain valuable for schedule design, and our result does not refute Lipschitz-guided schedules: in the tested systems the averaged-regularity *ordering* does not determine the fixed-NFE W_2 ordering, because fixed-budget error lives in solver-stage samples and transported signed defects the average cannot see. Every numerical statement carries a checksummed identifier from the anonymized audit bundle accompanying submission.

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